

Jacob Iwaniw

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EDUCATION

Georgia Institute of Technology

B.S. Mechanical Engineering, Spanish Minor | **GPA: 4.0/4.0**

Atlanta, GA

Expected May, 2029

Activities & Involvement:

- Ramblin' Rocket Club (GNC Structures Subteam), ASME, SAME, Beta Theta Pi Fraternity, Club Lacrosse

EXPERIENCE

Ramblin' Rocket Club

Atlanta, GA

Guidance, Navigation, and Control - Structures Subteam Member

Aug 2026 – Present

- Designed and iterated a SOLIDWORKS transition assembly connecting the 4-inch avionics section to the 6-inch motor and fin-actuation section of a fin-tab rocket.
- Sized load-bearing components using MATLAB-based stress, buckling, tear-out, and deflection calculations, achieving positive margins of safety across all components.
- Validated multiple structural load paths through ANSYS static structural FEA and refined component geometry, materials, fasteners, and 3D-printed parts based on simulated stresses.

Shattuck Labs

Research Triangle Park, NC

Technical Operations Engineering Intern

June 2026 – Aug 2026

- Developed a Python image analysis pipeline, layering a CNN with a YOLO object-detection model in Python to automate cell culture monoclonality verification from microscope imagery, reducing manual verification time from 20+ hours to 2 hours per batch.
- Trained and validated the model against a dataset of 2,000+ labeled microscope images captured via Cell Metric and VIPs imaging systems, achieving 90+% accuracy across a 1000 well dataset.
- Supported five production runs through sterile material preparation, cell aliquoting, and real-time monitoring and feeding of CHO-cell bioreactors up to 200 L.

Radiaid

Raleigh, NC

Co-Founder & Chief Engineer

March 2025 – Present

- Led development of an AI model for osteosarcoma malignancy classification from X-rays, achieving 93.75% accuracy on a dataset of 700+ images.
- Engineered the data preprocessing and augmentation pipeline to handle variable-quality radiographs, reducing overfitting and improving model robustness for clinics in low-resource settings.
- Secured patent-pending status for the core screening process through Duke University's Office of Translation and Commercialization.
- Toshiba Exploravision Science Competition Regional Winner, MIT Inventteams National Finalist, Samsung Solve for Tomorrow Winner, DECA Startup Pitch Competition Winner. (www.radiaidai.com)

DeAP Learning Labs

Raleigh, NC

Founding Engineer

April 2023 – Aug 2024

- Built AI chatbots to tutor high school students for AP classes/exams and provide feedback on their essays.
- Helped scale to 120,000 users, 2,000,000 questions answered, \$1M valuation, and Microsoft/OpenAI grants through backend prompt engineering. (<https://www.deaplearning.com/>)

PROJECTS

Loops - AR Transcription + Translation Glasses

February 2024

- Designed a pair of Augmented Reality glasses that provide the user with live speech-to-text transcription and live language translation in 25+ languages; Integrated an Arduino microcontroller and OLED display into a custom 3D-printed housing.
- Awarded "Best Rookie Team" at NC State Hackathon. (<https://devpost.com/software/loops>)

ADDITIONAL

Awards: USA Lacrosse Academic All-American, National Merit Scholarship Winner, Coca-Cola Scholars Program Semifinalist

Technical Skills: SOLIDWORKS, ANSYS Workbench, MATLAB, Python, PyTorch/Tensorflow, 3D printing, Laboratory documentation, Hand & Machining Tools, Microsoft Office.

Certifications: Certified Python Associate Programmer (PCAP), Certified SOLIDWORKS Associate (CSWA) (In Progress)

Languages: Fluent in English; Conversational in Spanish